UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

Imaging of pulmonary infection

Radiology department

4th stage 2026–2027

Assist. Prof Dr. Hiba Mohammed Abdulwahid

Radiology Specialist

Learning Objectives

1. **Describe** the common imaging modalities used in the evaluation of pulmonary infections,.
2. **Recognize** the characteristic imaging features of common pulmonary infections, including pneumonia, tuberculosis, and lung abscess.
3. **Differentiate** the main radiological patterns of pulmonary infections, including lobar consolidation, bronchopneumonia, and interstitial opacities.
4. **Identify** imaging features suggestive of complications of pulmonary infections, including cavitation, pleural effusion, and empyema

Imaging modalities used :

- Chest X ray
- Ct scan

General Imaging appearance of pulmonary infection :

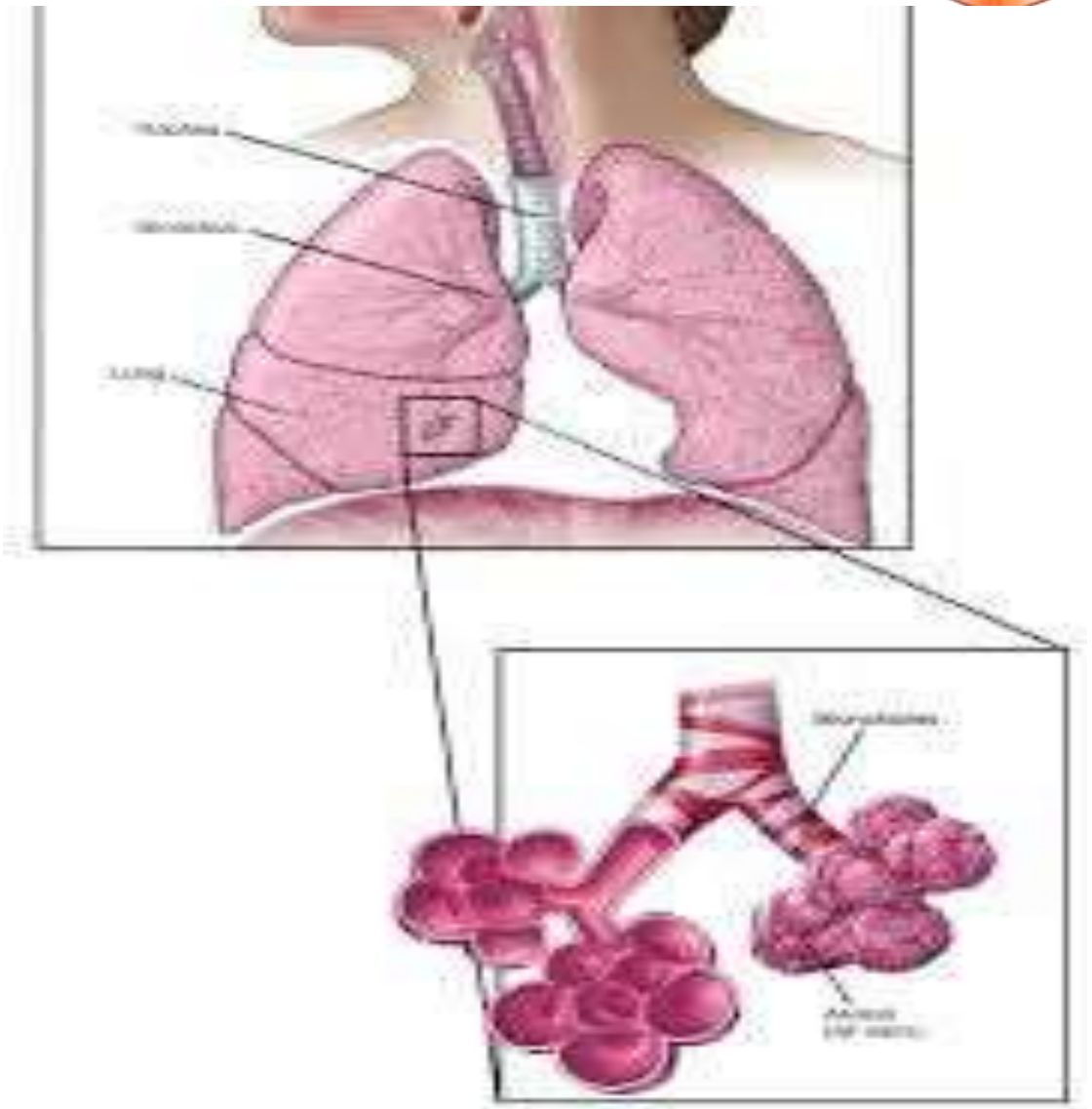
- Consolidation.
- Air bronchogram sign.
- Ground glass opacities
- Tree in bud.
- Cavitation

Mechanism of consolidation

Consolidation occurs through accumulation of inflammatory cellular exudate in the alveoli and alveolar ducts.

Radiological imaging , it presents as increased attenuation of the lung parenchyma causing obscuration of pulmonary vessels, without significant loss of volume

Air bronchogram can also be found



Consolidation

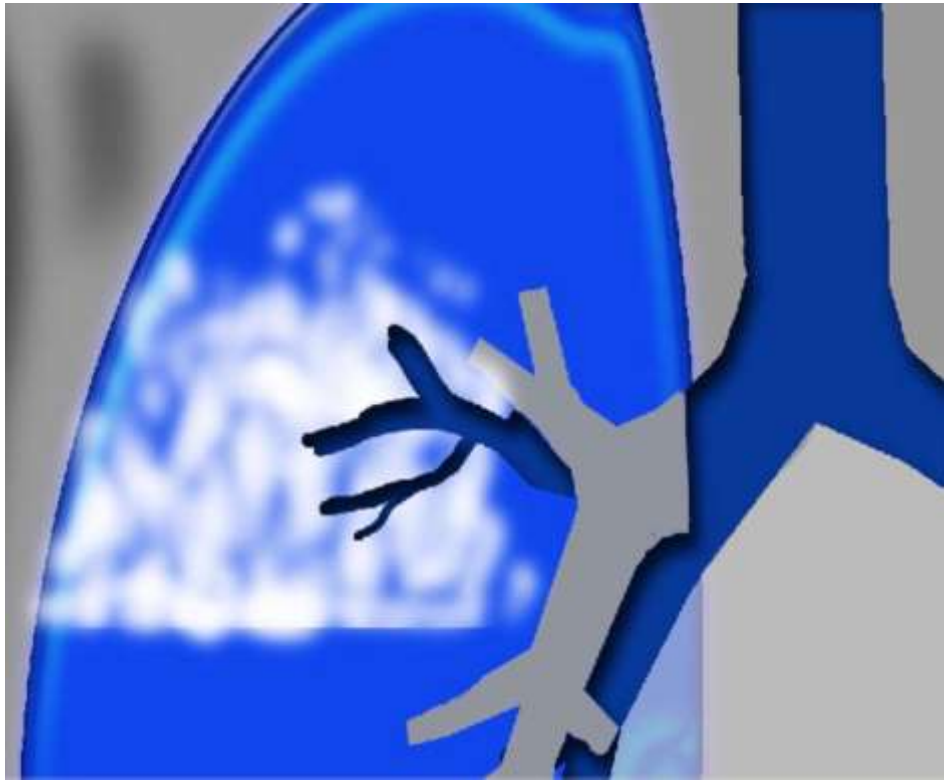
Replacement of air in the alveoli by other materials

- **Pus** pneumonia
- **Fluid** pulmonary edema
- **Blood** pulmonary hemorrhage
- **Cells** bronchoalveolar carcinoma, lymphoma, and chronic infection



Consolidation

- **ill-defined or well defined**
- **air-bronchogram**
- **NO** volume loss

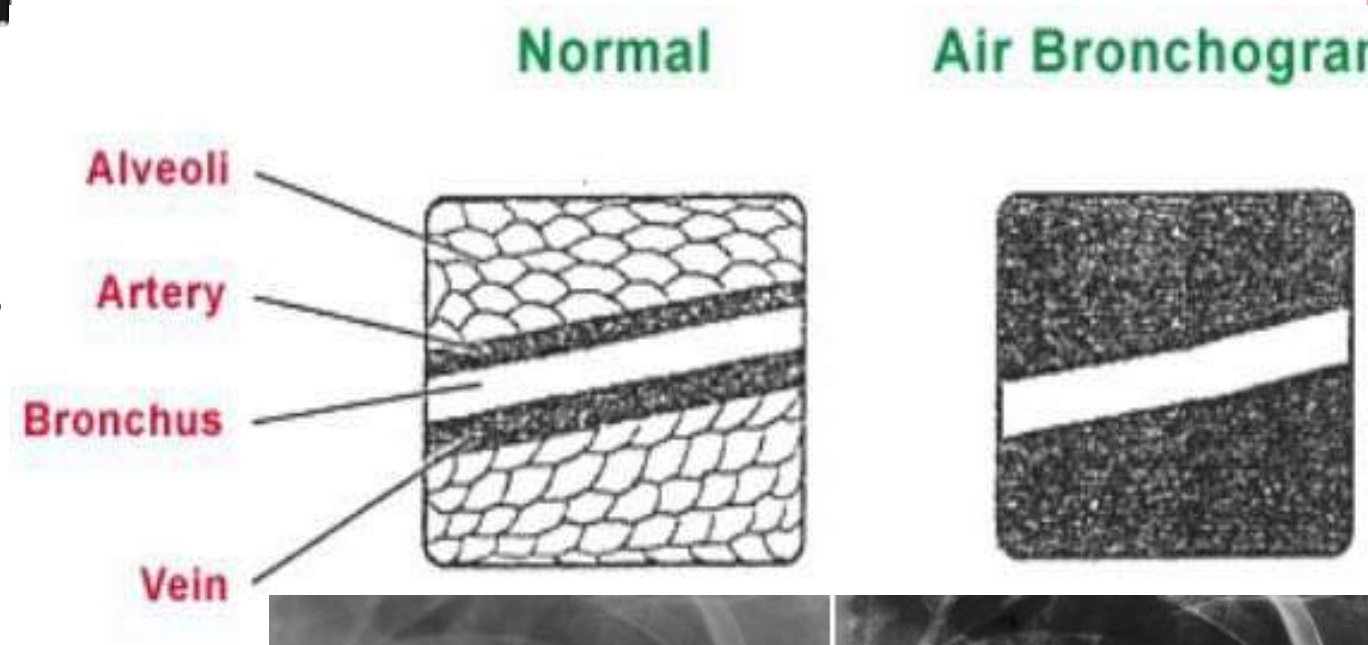


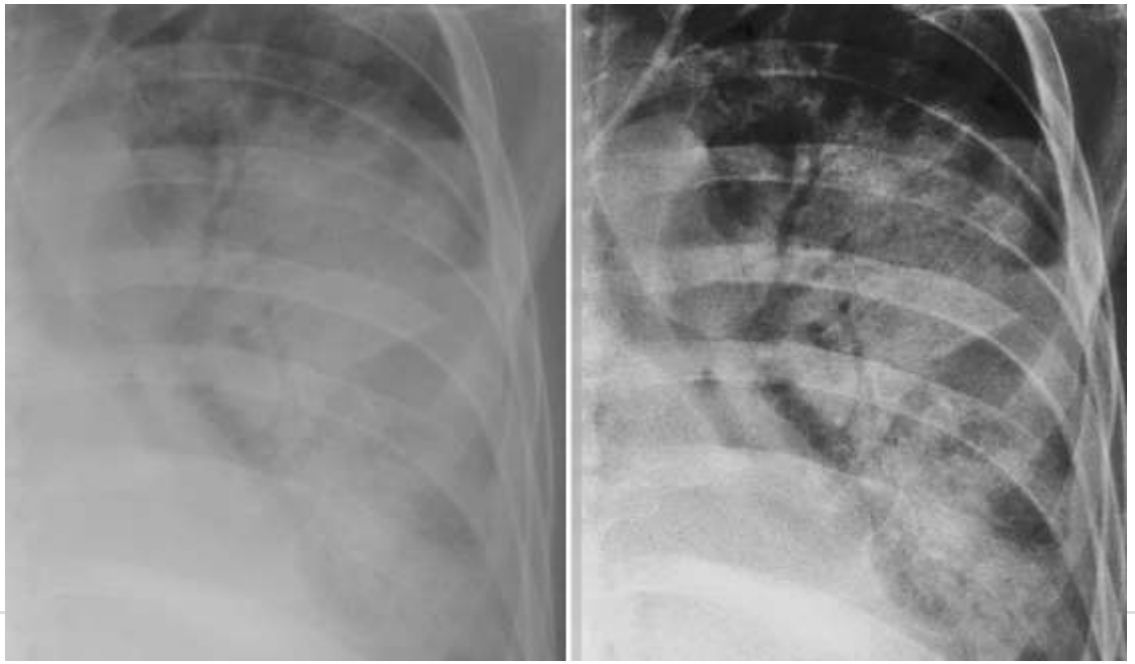


Consolidation

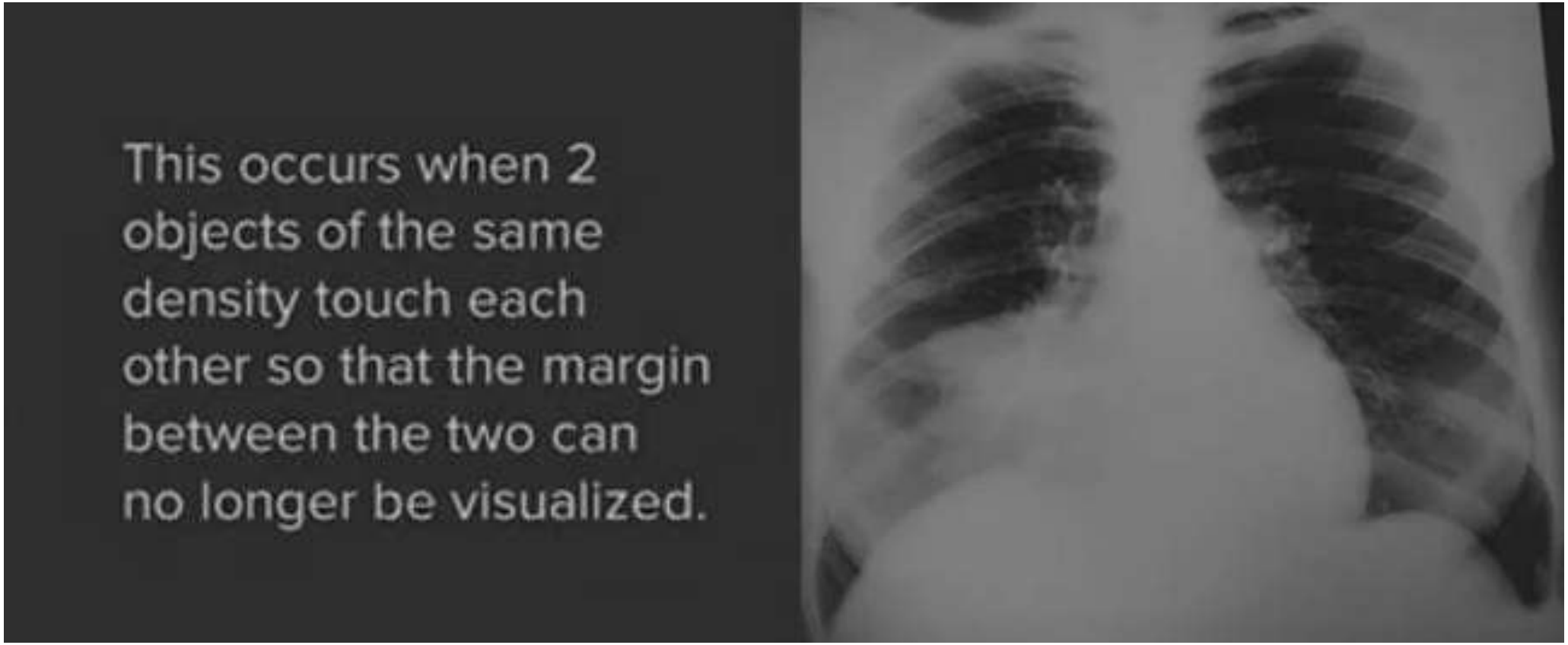
Air bronchogram

Air in the bronchiole appear as lucencies against surrounding opacification



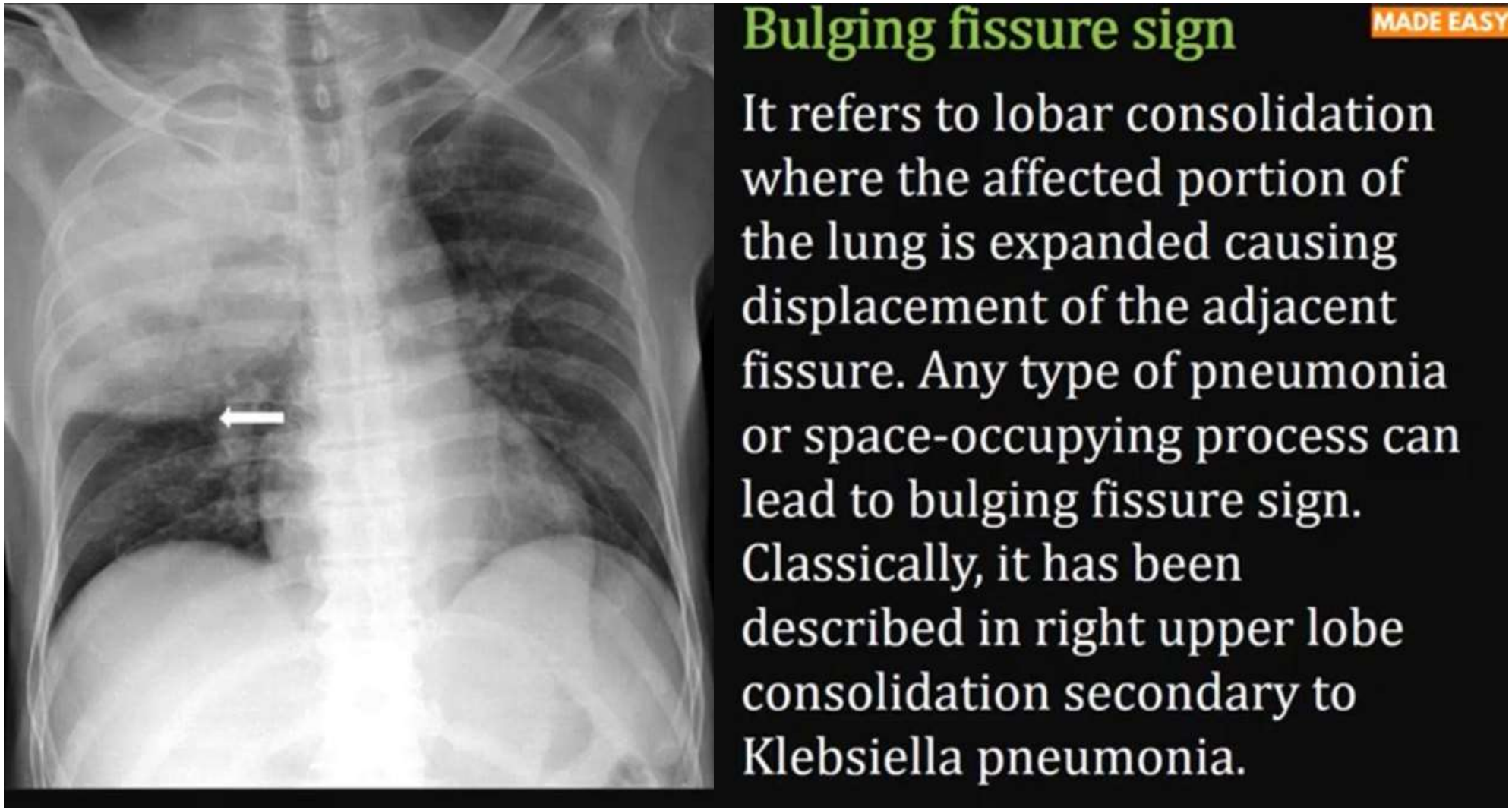


Silhouette sign

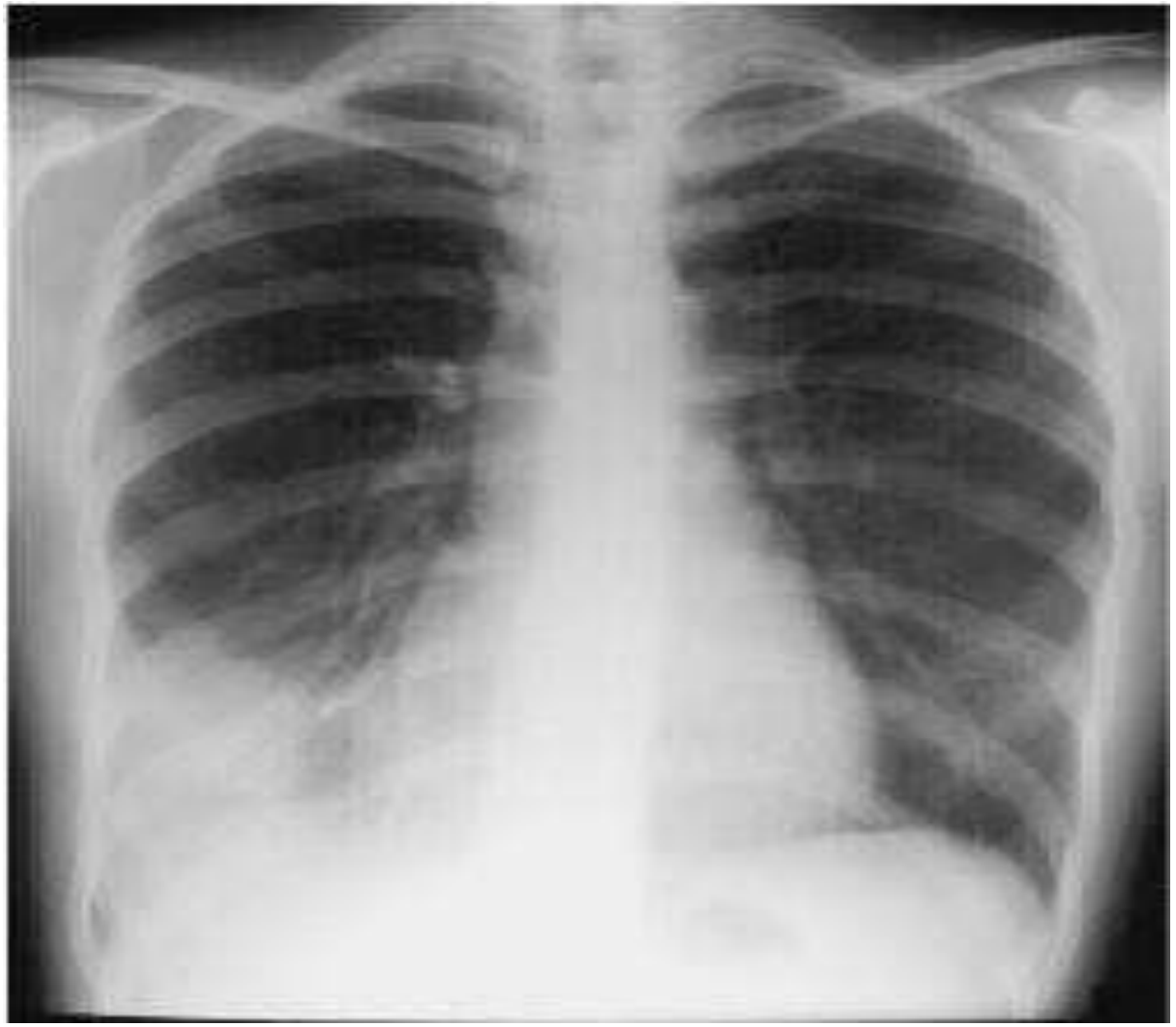


Silhouette sign

Silhouette / Structure	Contact with Lung
Upper right heart border/ascending aorta	Anterior segment of RUL
Right heart border	RML (medial)
Upper left heart border	Anterior segment of LUL
Left heart border	Lingula (anterior)
Aortic knob	Apical portion of LUL (posterior)
Anterior hemidiaphragms	Lower lobes (anterior)


Bulging fissure sign
MADE EASY
It refers to lobar consolidation where the affected portion of the lung is expanded causing displacement of the adjacent fissure. Any type of pneumonia or space-occupying process can lead to bulging fissure sign. Classically, it has been described in right upper lobe consolidation secondary to Klebsiella pneumonia.

Case



- Case



Bacterial Pneumonia

Radiological features of pneumonia

Lobar Consolidation :

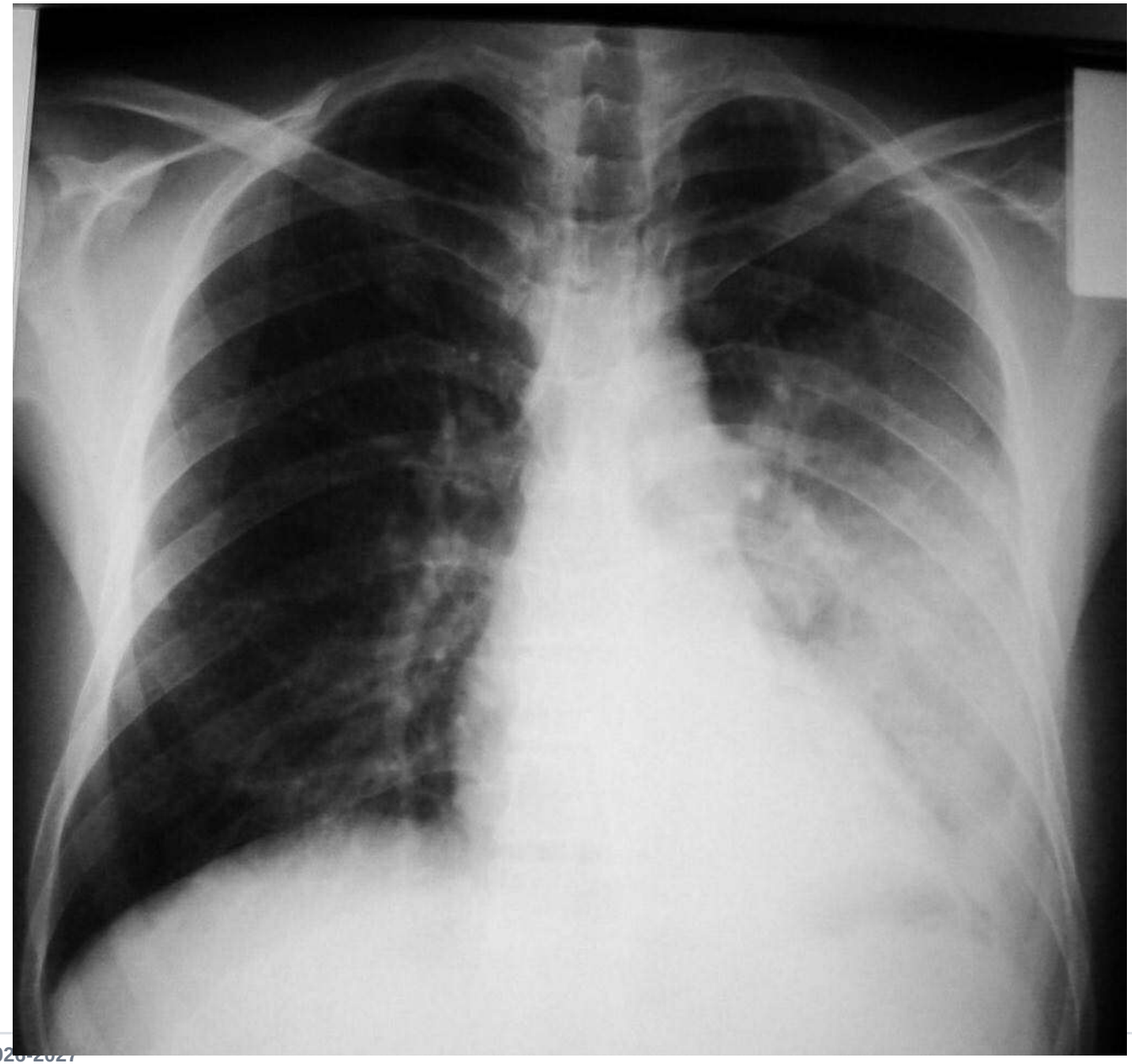
- **Homogenous opacification in lobar pattern.**
- **Sharply defined margin at the fissure.**
- **Air-bronchogram.**



The cause of lobar consolidation is

- streptococcal
- staphylococcal
- Klebsialla.

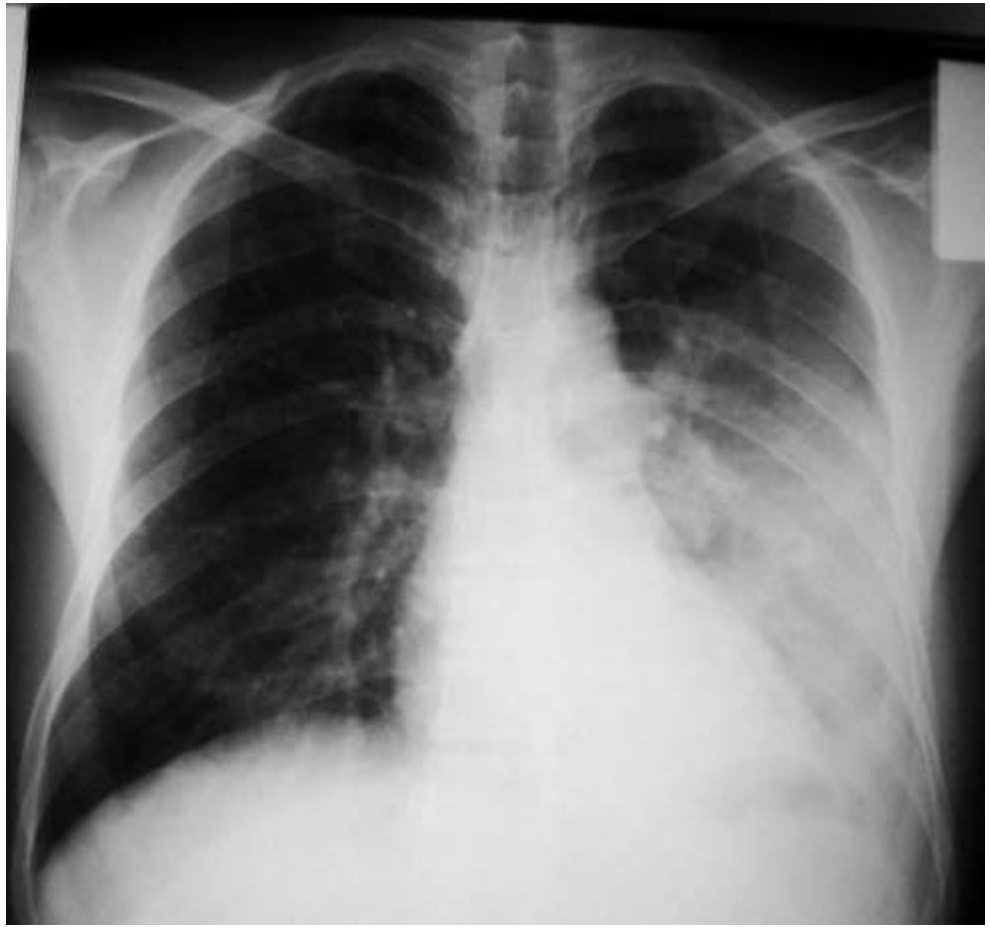
Case



Answer

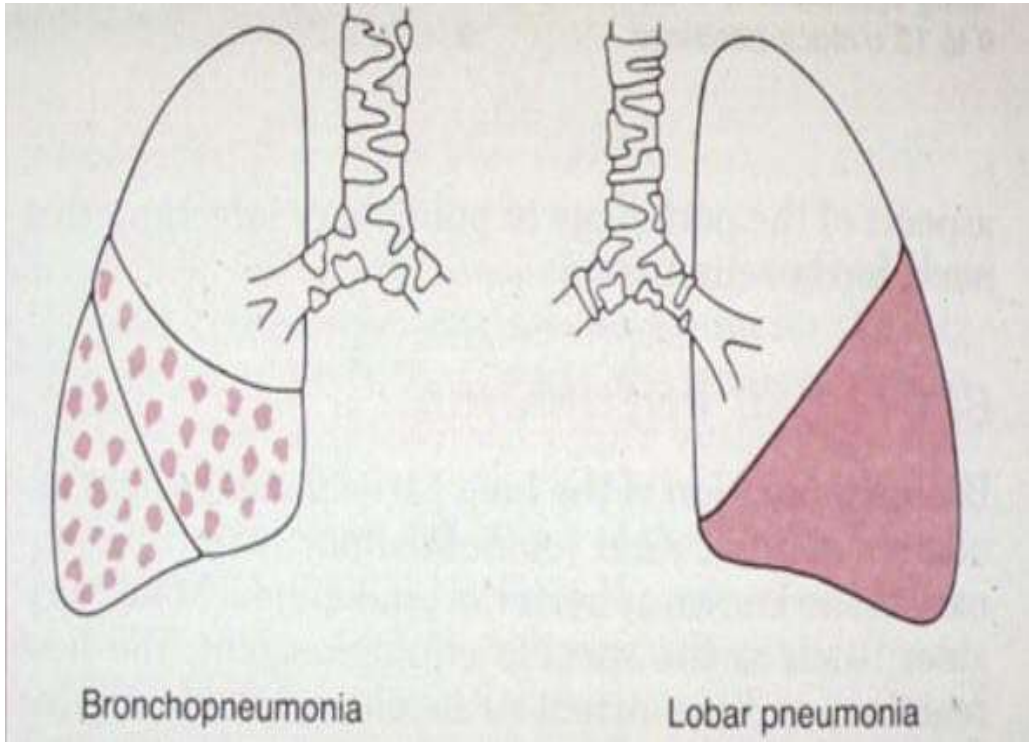
- Opacification of left lower and middle zone silhouetting left hemidiaphragm

- Dx: Left lower lobe consolidation



2.Bronchopneumonia:

- When the consolidation is **patchy** involve one or more lobes (**multilobar**)
- Multiple **nodular** or **reticulonodular** opacities
- **Bilateral a**symmetrical.

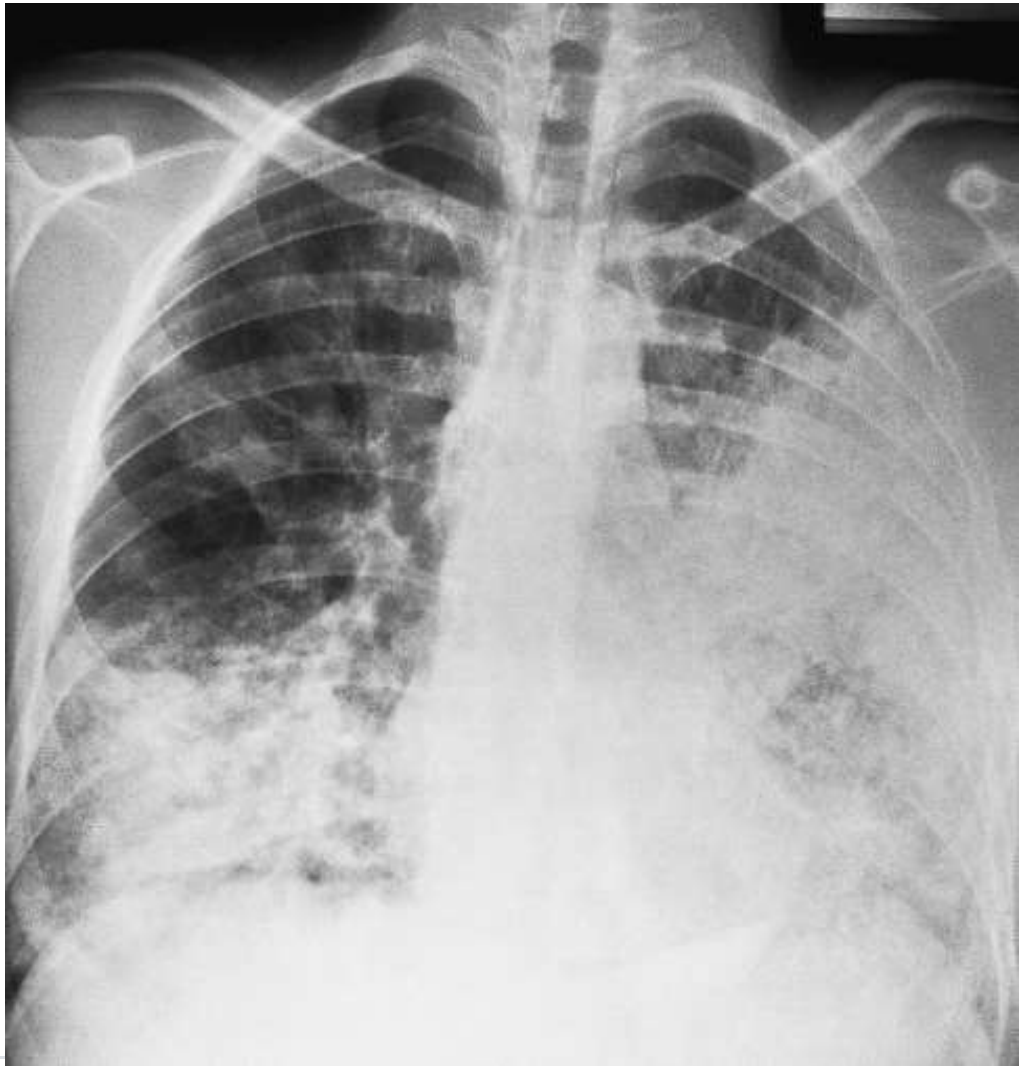


The cause of bronchopneumonia:

- Staph aureus
- Gram _ve

Bronchopneumonia:

- Bilateral asymmetrical patchy consolidation



Round pneumonia

Seen in pediatric patients.

- Well defined or ill defined opacity.
- The cause of round pneumonia is **streptococcus**



Viral and mycoplasma pneumonia

- Wide spread ill-defined consolidation.
- Loss of clarity of vascular markings.



Cavitation

- What are the differential diagnosis of pulmonary cavity ?

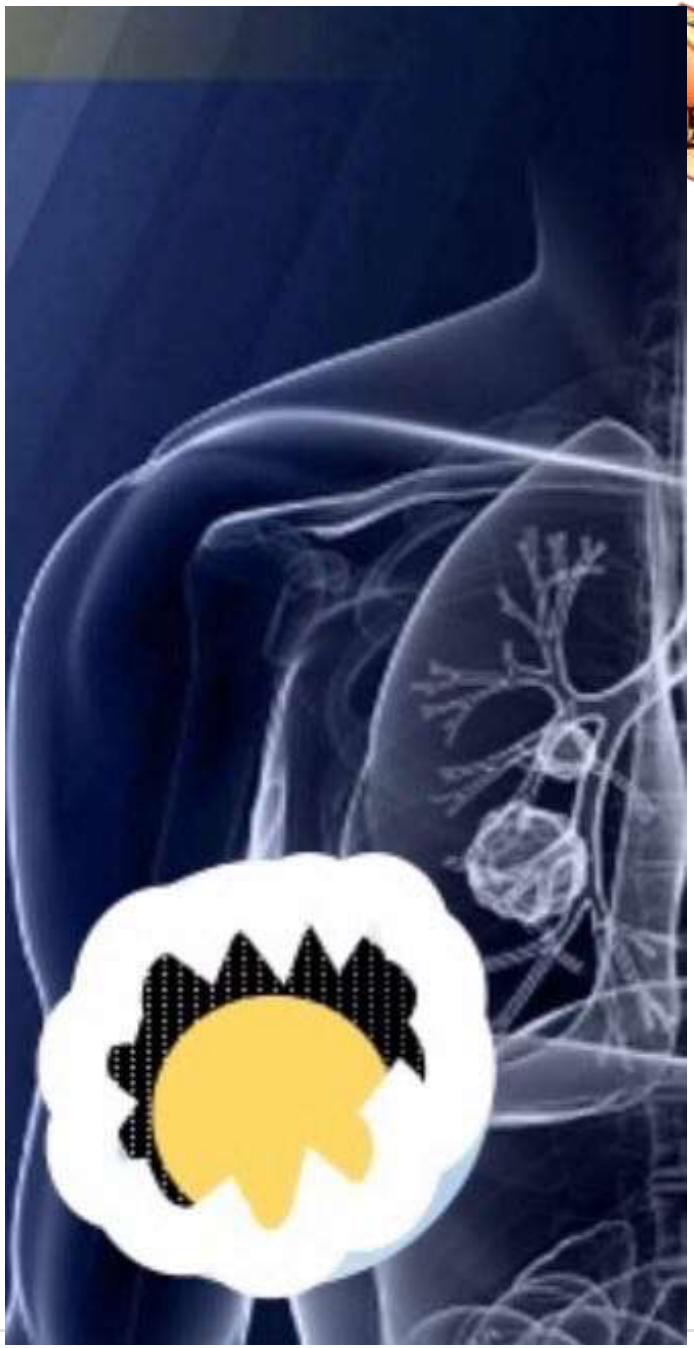


Cavitation

Differential diagnosis of cavitary lung lesion (important)

- Lung abscess.
- Post-primary TB.
- Cavitating neoplasm
- Wegener's granulomatosis.
- Infected emboli

How to distinguish between malignant cavity and benign cavity ?



Cavitation

ng lung neoplasm or cavitation in
uses.

Cavity morphology

- Thickness of the cavity wall
- The character of its inner lining (whether irregular or smooth)
- The presence and nature of its contents



Morphology Of The Cavity

Benign cavity

- Thin wall<5mm (92 % benign)
- Smooth inner wall

Malignant cavity

- Thick wall> 15mm (95% malignant)
- Nodular inner wall

Content Of The Cavity

Non specific

Flat smooth air fluid level

- **Benign ; abscess**
- **Necrotic cancer**

More specific

Mobile intra-cavitary lesion

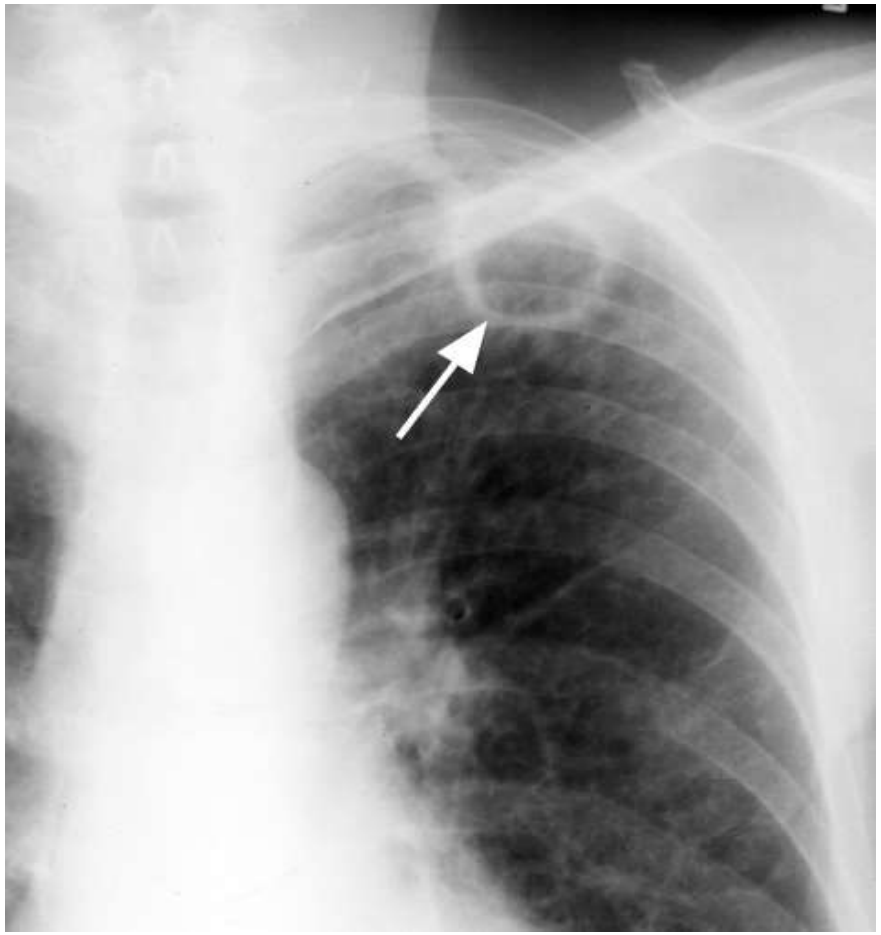
- **Fungus ball.**
- **Blood clot**

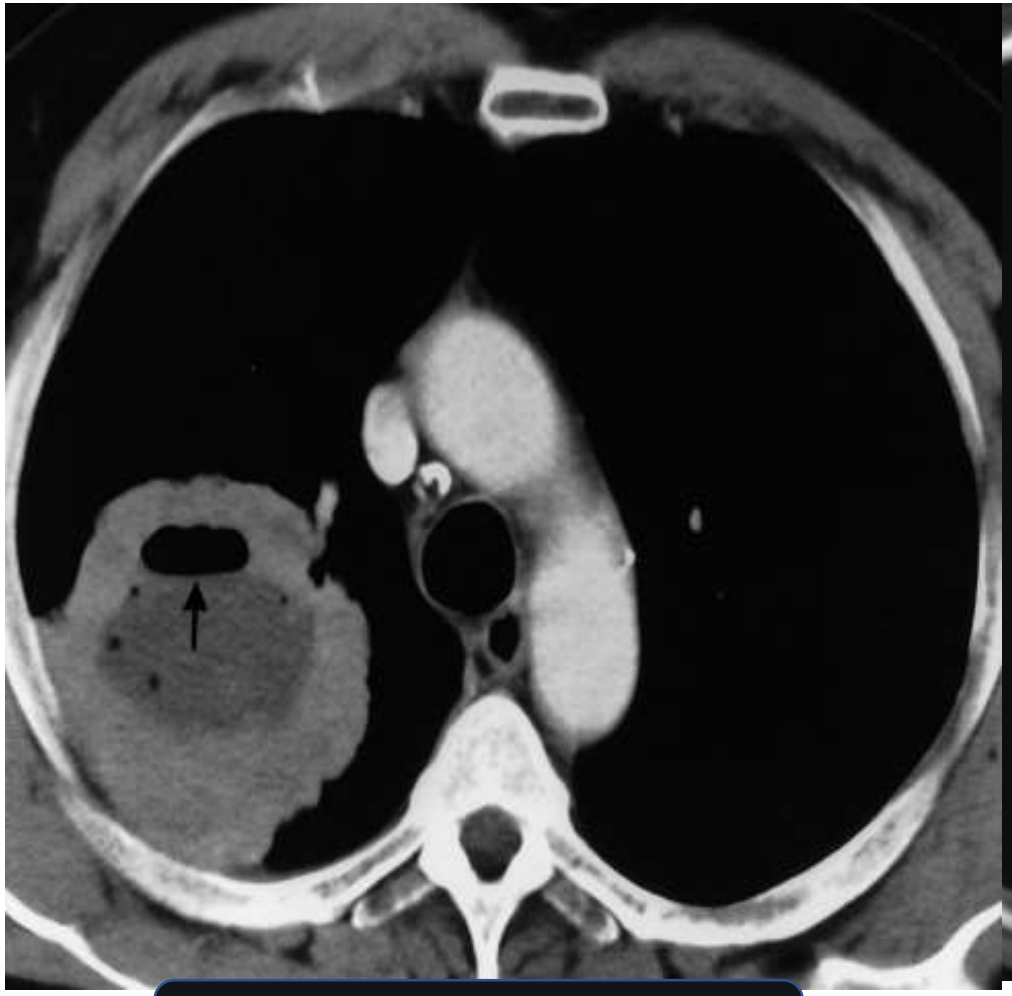
Collapsed membrane

- **Ruptured Hydatid**

Benign lung cavity

Well defined thin walled cavity in left upper zone
No fluid level
Clear surrounding lung





Lung cancer

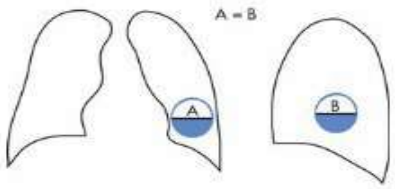
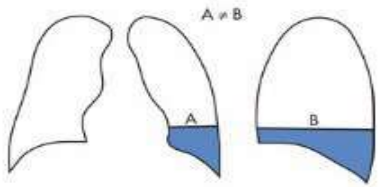


Lung abscess with adjacent consolidation

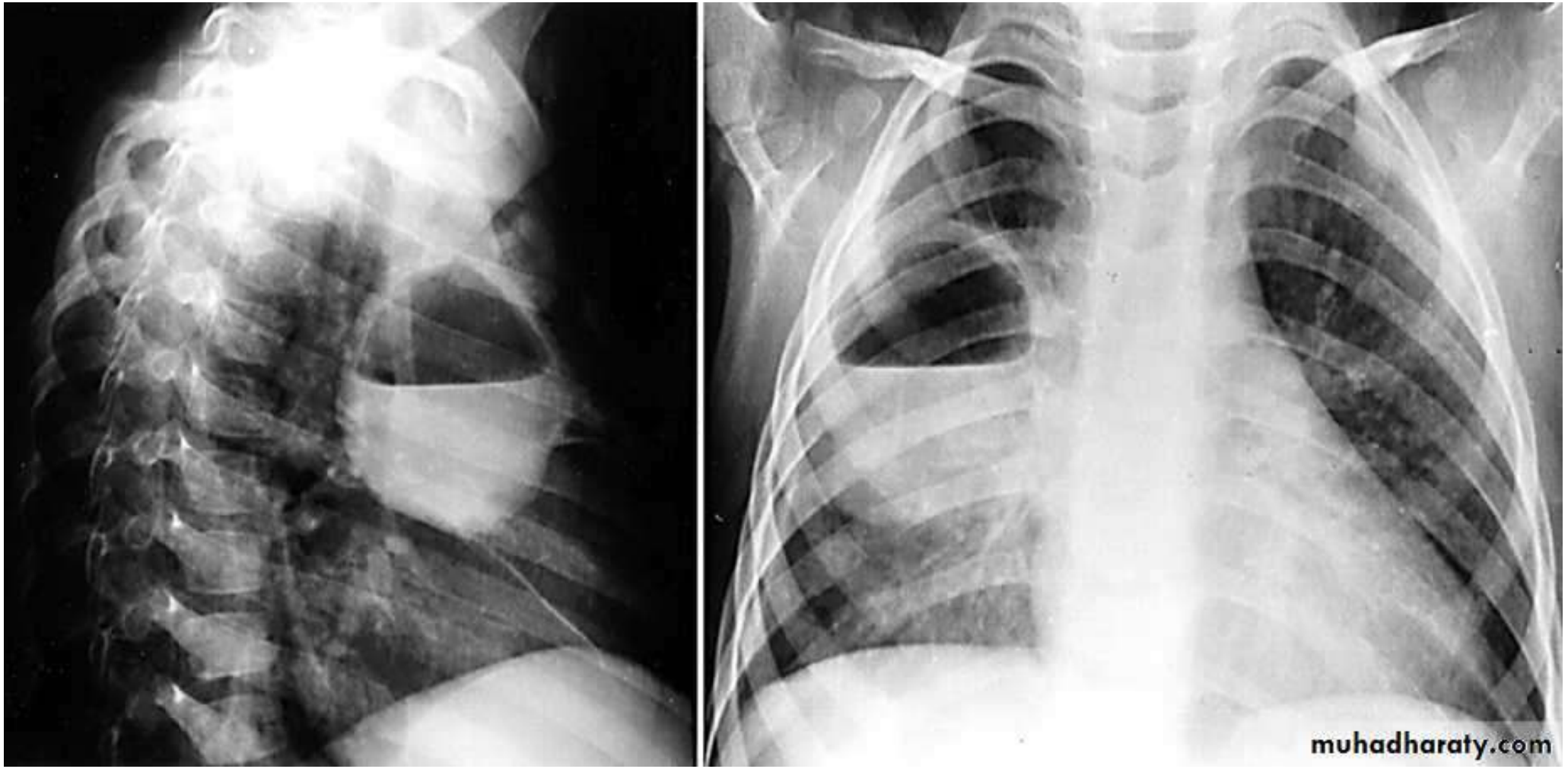
Lung Abscess

- Localized suppurative lesion of lung parenchyma.
- Similar shape on PA and lateral view

Lung abscess VS empyema

	Abscess	Empyema
Cause	Necrotizing pneumonia (anaerobes, fungus)	Abscess extends to pleura; trauma, surgery
Shape	Round	Elliptical along chest wall
Air-fluid level	A = B	A ≠ B
Margins	Sharp or irregular	Sharp
Wall	Thick	Thin
Lung	Normal position	Displaced
Pleura	Not seen	Split
Vessel/bronchi	Within	Displaced
Treatment	Antibiotics, postural drainage, percutaneous drainage in nonresponders	Percutaneous drainage
		

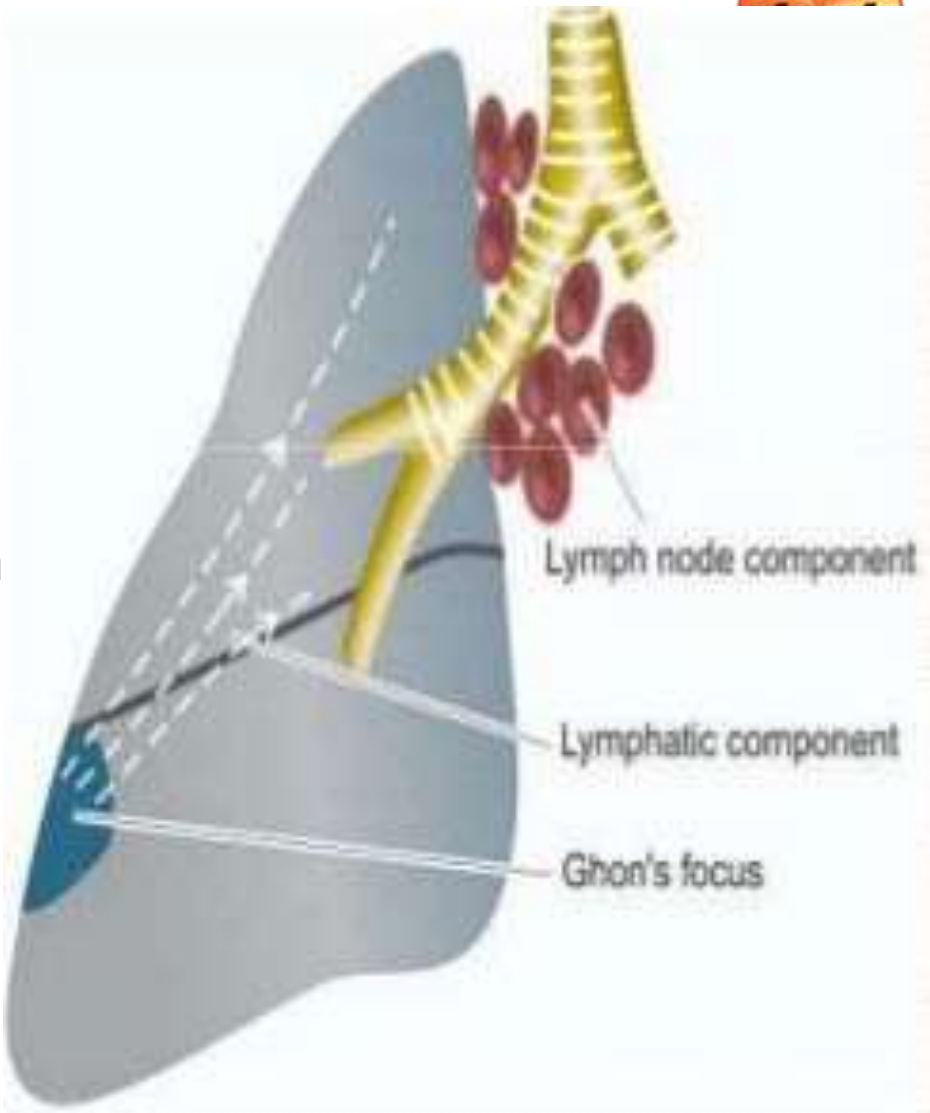
Lung Abscess



Primary Pulmonary tuberculosis

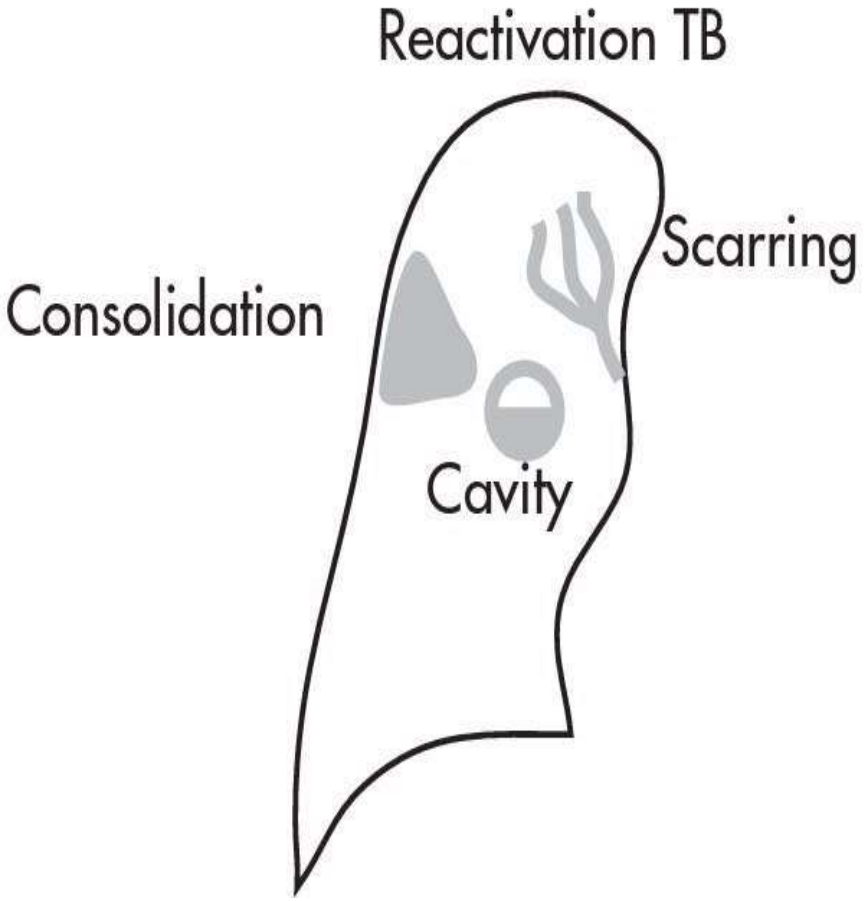
Mycobacterium tuberculosis

- **Ghon focus :** area of consolidation, in the periphery the lung– usually in the mid or upper zones.
- Often **enlarged** hilar and contiguous mediastinal (pa tracheal) lymph nodes (usually right sided).
- **primary complex** = pulmonary consolidation **+** lymphadenopathy



Post-primary tuberculosis

- The usual form in **adults**, is believed to be re-infection,
- Usually confined to the **upper posterior** portions of the chest or apical segment of lower lobe.
- Multiple small areas of consolidation and are often bilateral.
- frequently **cavitate**.
- Healing occurs by **fibrosis**, often with **calcification**



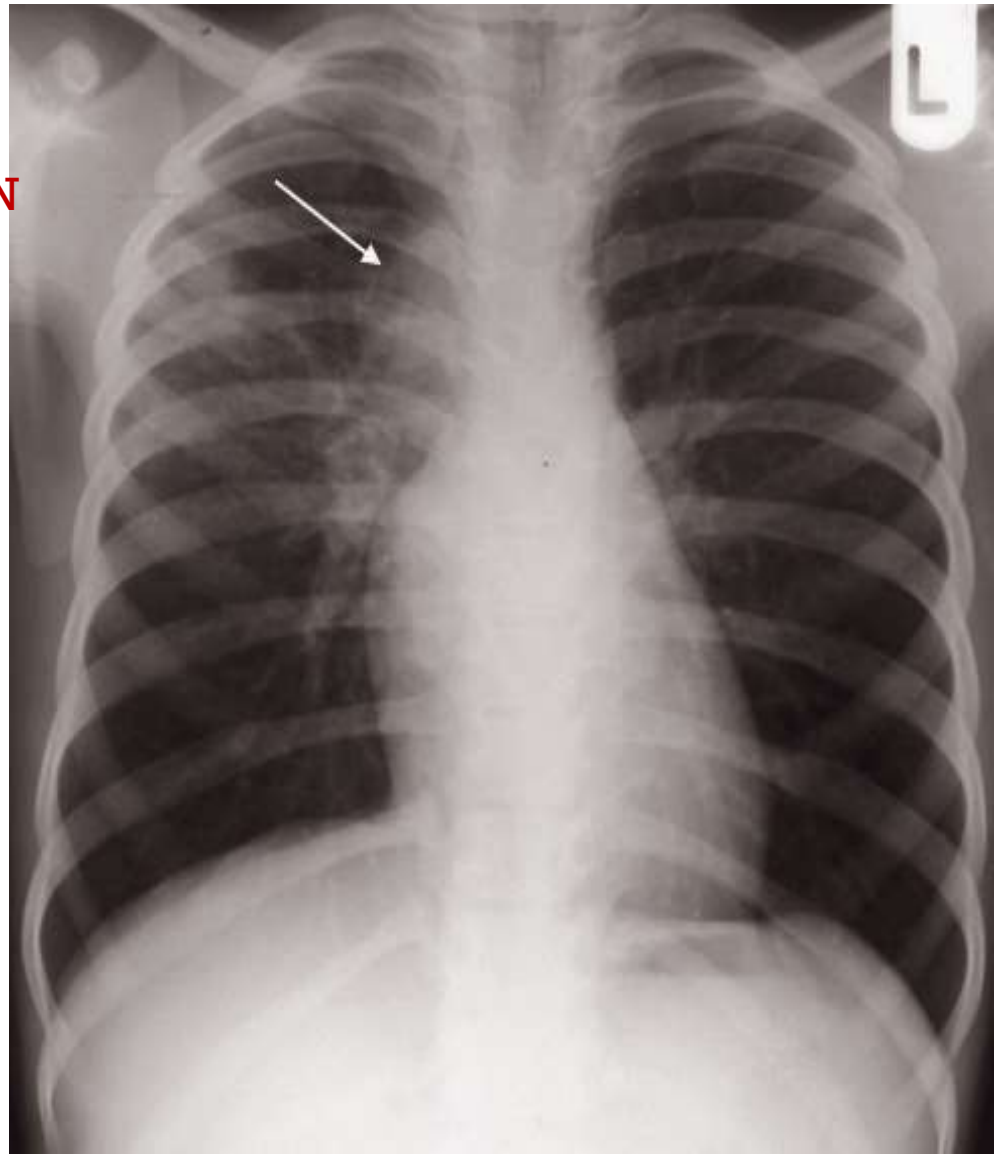
	Primary TB	Post-primary TB
Location	No lobar predilection	Upper lobe posterior segment or apical segment of lower lobe
appearance	Usually focal	Multifocal patchy involvement
cavitation	No	Frequent(indicate active disease)
LAP	Common	Typically absent
Miliary pattern	yes	yes
Tree in bud	Uncommon	Common (indicate active disease)

Primary Pulmonary tuberculosis

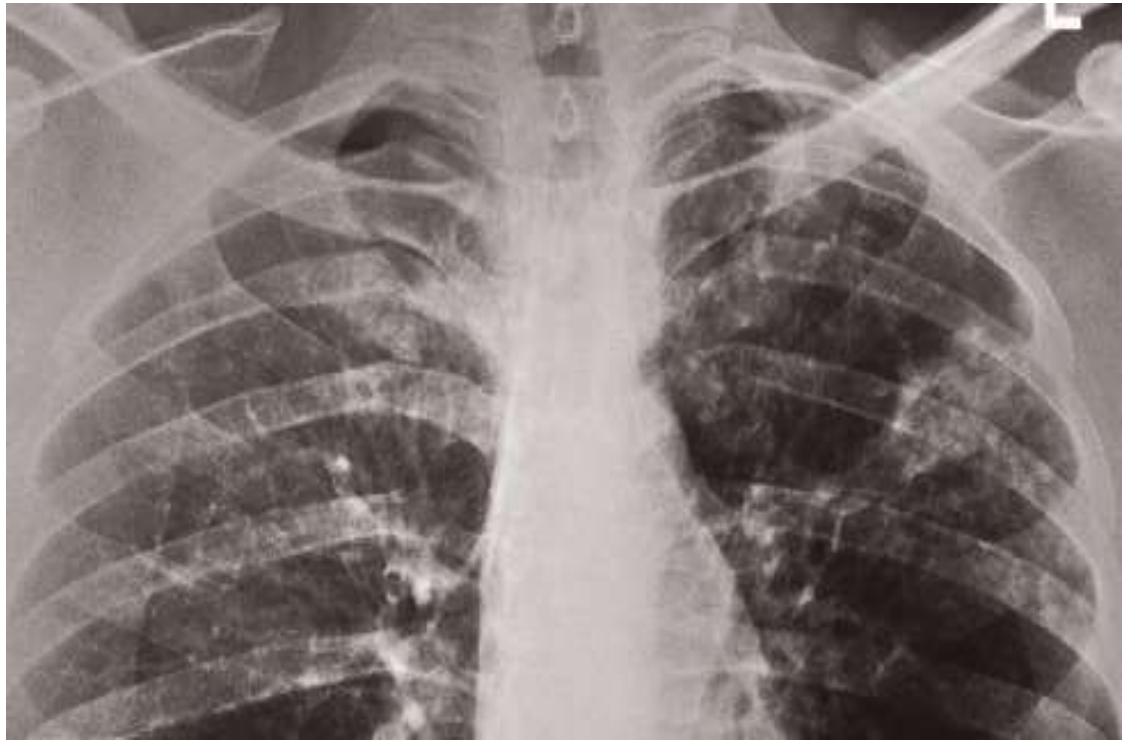


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para-tracheal LN

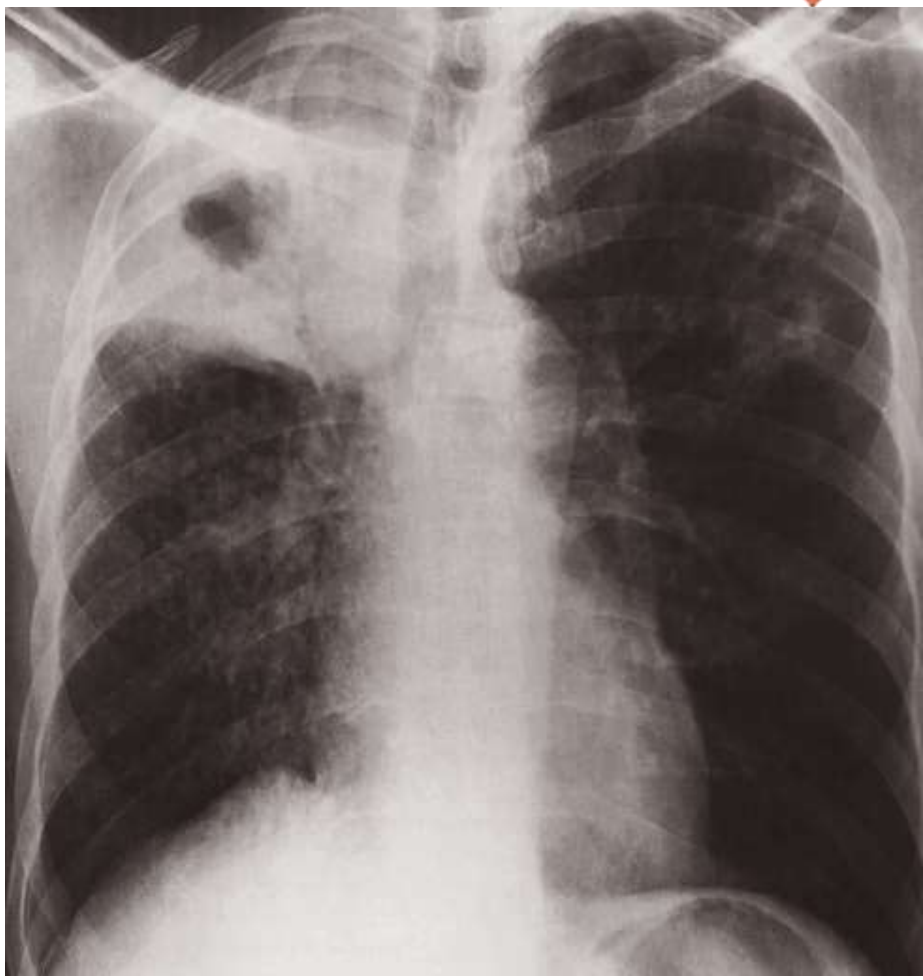


- ill-defined consolidations scattered in both upper lobes.
- Dx : post-primary TB



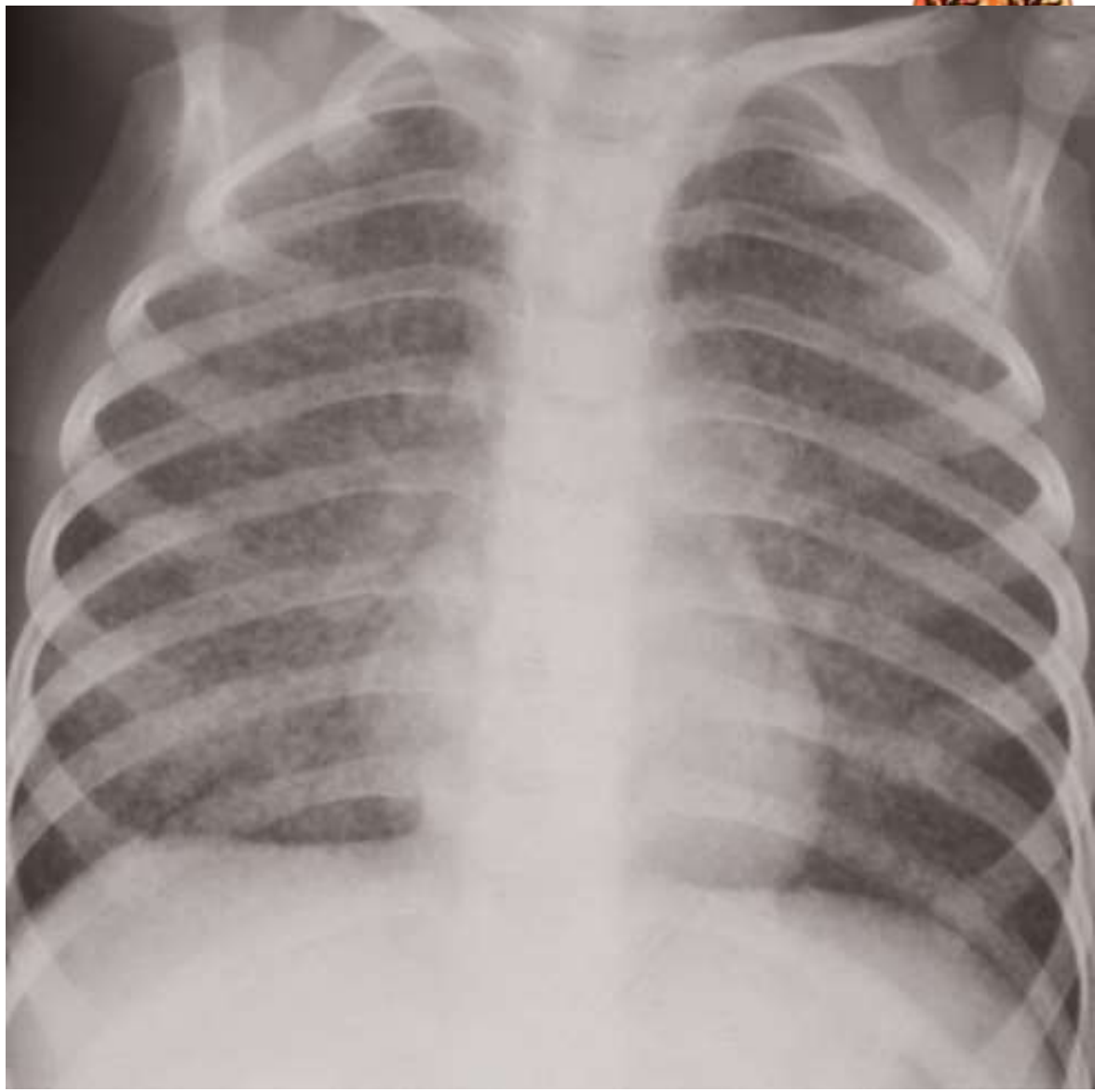
- Right upper lobe is consolidated and contains a large central cavity

- Dx : post-primary TB
- Ddx :



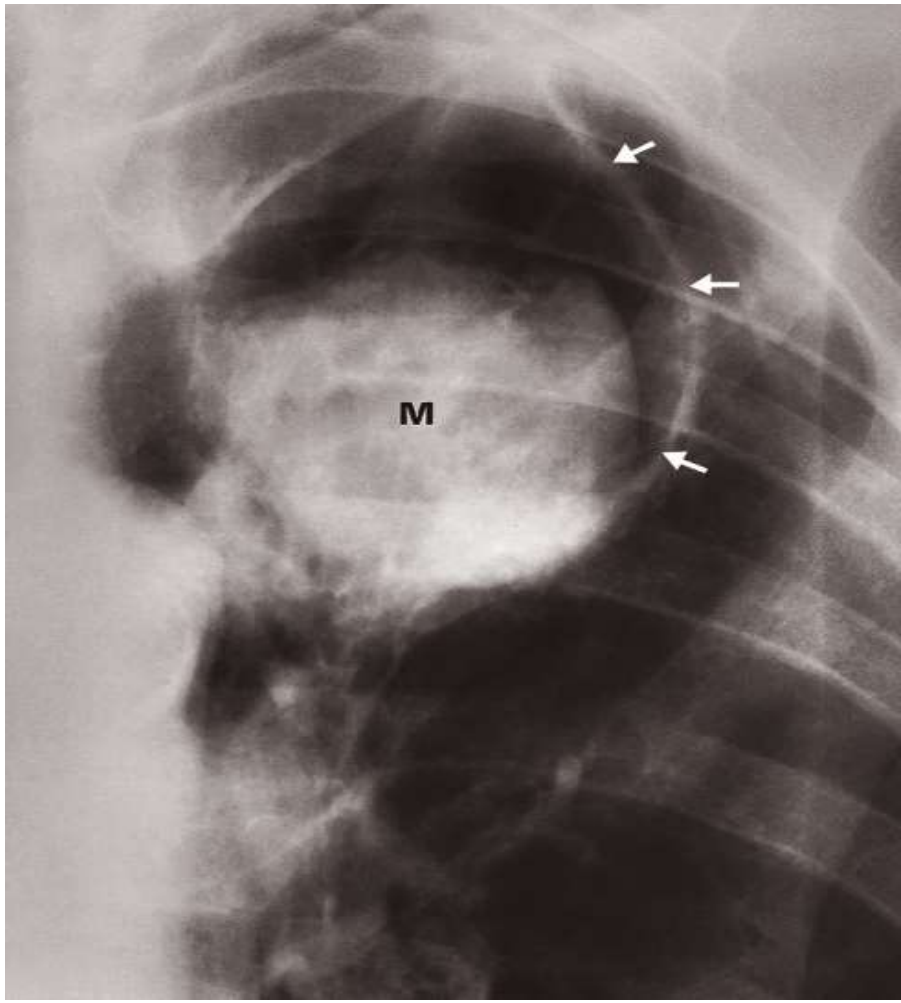
Miliary TB

- Due to hematogenous spread of infection.
- Occur in primary or post-primary TB.
- **Tiny (1-2mm)nodules** distributed in all lung zones
- evenly distributed throughout both lungs.
- carries a **poor prognosis**



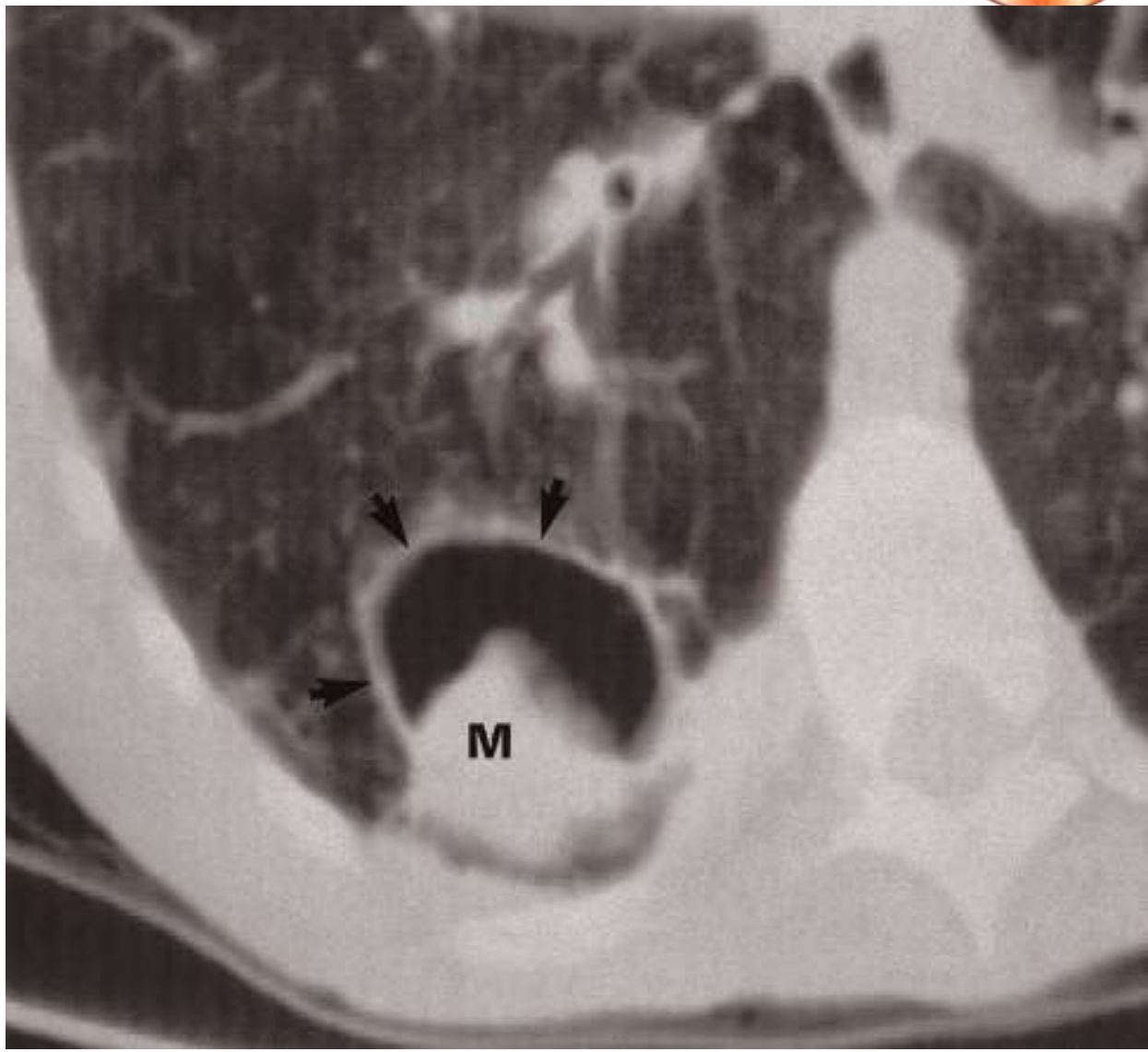
Mycetoma

- Aspergillus fumigatus may colonize cavities (e.g. TB cavity)
- produce a **ball of fungus** (Mycetoma)
- Air is seen between Mycetoma and wall of cavity (air cresent sign).
- These cavities are surrounded by evidence of old TB **(fibrosis and calcification).**



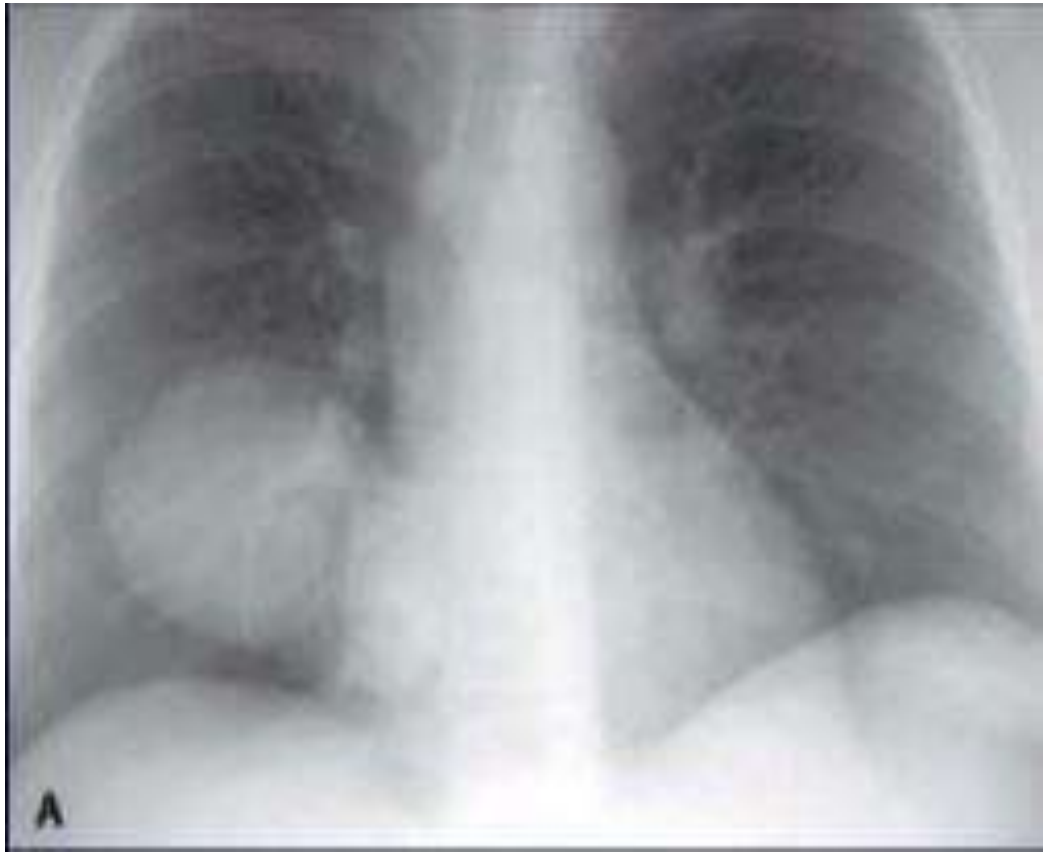
Case

- Modality.
- Findings
- Dx.



Hydatid Disease

- cysts in the lung or pleural cavity
- Solitary or multiple.
- spherical shadows **with very well-defined borders.**
- occasionally rupture resulting in **water lily appearance**.





CT scan of chest (lung and mediastinal window)

There is a well defined non enhancing fluid density lesion (cystic lesion)

DX hydatid cyst


thank
you